TORRENS UNIVERSITY AUSTRALIA

Assessment 2: ICT Governance and Policy Analysis

Case Study: United Parcel Service (UPS)

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1. Introduction

Ethical theories form the philosophical backbone of ICT governance, shaping how organisations design, deploy, and oversee technology systems. Utilitarianism holds that IT decisions should maximise overall benefit across all stakeholders, including employees, customers, and the public. Deontological ethics demands that organisations fulfil their duties and honour obligations regardless of commercial outcomes, such as protecting user privacy even when doing so is costly. Virtue ethics shifts focus to the character of decision-makers, encouraging the cultivation of responsible, transparent, and fair practices over time (Johnson and Krauss, 2020). In the context of IT, these theories translate into concrete governance obligations: accountability, transparency, security, and alignment with societal values.

IT governance is the framework of structures, processes, and relational mechanisms through which organisations direct and control their technology resources to achieve strategic objectives. Effective governance ensures that IT investments generate measurable value, that risks are proactively managed, and that technology activities remain aligned with business goals (Weill and Ross, 2004). Organisations that neglect IT governance expose themselves to competitive displacement, operational failure, and reputational harm.

United Parcel Service (UPS) referred to in this case study as Universal Parcel Services is one of the world's largest package delivery and supply chain management companies, operating in over 220 countries. Founded in 1907, UPS faced a serious competitive threat in the mid-1980s when FedEx's superior tracking technology began eroding its market share in air services. This report analyses UPS's ICT governance policy, evaluates its alignment with established international standards, and critically assesses its ethical dimensions, before offering targeted recommendations for improvement.

2. Body

2.1 UPS ICT Policy

When Oz Nelson became CEO of UPS in 1986, senior management committed $11 billion over a decade to a comprehensive ICT transformation. This included building a state-of-the-art data centre, establishing a global network, developing shared databases, implementing enterprise-wide applications, and constructing a redundant disaster-recovery environment (UPS Case Study, 2024). Critically, UPS recognised that technology investment alone was insufficient; governance processes were needed to ensure that spending targeted key business objectives.

UPS's ICT governance structure operated through four interconnected layers, as illustrated in Table 1 below. The Senior Management IT Steering Committee set strategic direction and approved major investments, mandating a highly centralised and standardised IT environment to ensure reliability and consistent customer service. Beneath this, the IT-PAX Committee, led by the CIO, enforced architecture standards while preserving the flexibility required as UPS diversified into supply chain financing and service parts logistics. A dedicated Standards Committee, headed by the top IT architect, managed day-to-day technology decisions, escalating matters with strategic implications to the PAX or Steering Committee as required. Business leaders, not IT staff alone, participated actively through a project charter process, submitting proposals that documented expected costs and benefits before any investment was approved.

Table 1: UPS IT Governance Structure

Governance Layer	Governing Body	Key Responsibility	Decision Scope
Strategic	IT Steering Committee	Set IT direction; approve major investments; mandate standardisation	Enterprise-wide
Architectural	IT-PAX Committee (led by CIO)	Enforce architecture standards; balance standardisation with flexibility	IT architecture
Operational	Standards Committee (led by Top IT Architect)	Manage day-to-day technology standards; flag strategic implications	Application-level
Business	Core Process Teams (CRM, Package Mgmt, Product Mgmt)	Identify IT priorities; submit project charters with cost/benefit analysis	Business unit

This layered, multi-stakeholder model ensured that IT decisions at every level remained tethered to business strategy, operational requirements, and financial discipline.

2.2 Business and IT Alignment

Business and IT alignment refers to the degree to which IT strategy supports and enables business strategy (Henderson and Venkatraman, 1993). UPS's governance mechanisms demonstrate a high level of alignment. The project charter process required any proposed IT investment to articulate measurable costs and benefits, which prevented speculative spending and kept technology directly linked to value creation. The three cross-functional core processes identified by the executive team, namely customer relationship management, package management, and product management, provided a clear framework within which IT priorities were set by business leaders rather than technology specialists in isolation.

The benefits of this alignment were substantial and demonstrable. UPS reduced operating costs significantly through technology-driven efficiencies. Customer service quality improved through real-time package tracking and consistent data access. When the internet emerged as a strategic opportunity in the mid-1990s, UPS's existing governance mechanisms enabled key managers to recognise its importance quickly and act decisively, generating early competitive advantage in e-business (UPS Case Study, 2024). A potential drawback, however, was the risk of over-centralisation: rigid standardisation could limit the agility of individual business units, particularly those with specialised technology requirements.

2.3 Ethical Analysis of UPS ICT Strategy

From a utilitarian perspective, UPS's ICT strategy generated broad benefits: customers gained reliable package tracking and consistent service quality, employees benefited from stable and efficient systems, and shareholders received improved financial returns. Deontologically, the formalised governance structures reflect a commitment to procedural fairness, as investment

decisions followed documented, transparent processes rather than ad hoc executive preferences (Stahl, 2004).

However, a significant ethical concern arises from the concentration of customer data within a highly centralised infrastructure. Large shared databases containing personal tracking information create real risks around data privacy, surveillance, and potential misuse. The case study as described does not document explicit data ethics policies, consent frameworks, or mechanisms to protect customers from unauthorised use of their information. From a deontological standpoint, the absence of such safeguards represents a failure of duty to customers, regardless of whether any harm actually occurred. This gap carries increasing weight as data protection expectations and regulatory requirements have intensified considerably since the 1990s.

3. Recommendations

3.1 Comparison with ICT Governance Standards

Table 2 below provides a structured comparison of UPS's ICT policy against five established ICT governance frameworks. This comparison reveals both the strengths of UPS's approach and the areas requiring improvement.

Table 2: UPS ICT Policy vs. Established ICT Governance Standards

Framework	Alignment with UPS ICT Strategy	Gap / Recommendation
COBIT	Tiered governance (Steering, PAX, Standards) mirrors COBIT's EDM (Evaluate, Direct, Monitor) domains. CIO participation at all levels reflects board-level IT accountability.	No documented maturity assessment. Recommend formal COBIT maturity benchmarking to measure governance effectiveness.
ITIL	Standardised service delivery, reliability focus, and redundant disaster-recovery environment align with ITIL service continuity and availability management.	No evidence of formal ITIL service catalogue or incident management process. Formalising these would strengthen operational governance.
ISO 27001/002	Redundant operations environment indicates awareness of availability risks.	No formal Information Security Management System (ISMS) documented. ISO 27001 certification recommended to address data confidentiality risks.
COSO	Steering committee oversight and project charter approval process reflect COSO control environment and risk assessment components.	No explicit risk register or internal audit trail described. Formal enterprise risk management integration recommended.

Framework	Alignment with UPS ICT Strategy	Gap / Recommendation
ISO 9000	Architecture standardisation enforced by PAX Committee reflects ISO 9000 quality consistency principles.	Quality management not formally certified. Certification would strengthen external credibility and continual improvement culture.

As Table 2 demonstrates, UPS's governance model aligns strongly with the structural and service management dimensions of COBIT and ITIL, while exhibiting meaningful gaps in information security management (ISO 27001), formal risk management (COSO), and quality certification (ISO 9000). These gaps should be addressed systematically through the recommendations below

3.2 Specific Recommendations

- **Establish a formal Information Security Management System (ISMS) aligned with ISO 27001.** The centralised data infrastructure at UPS requires rigorous security governance. Pursuing ISO 27001 certification would formalise controls over data confidentiality, integrity, and availability, and provide customers with independent assurance that their information is protected.
- **Develop and embed a Data Ethics Framework.** UPS should implement clear policies governing the collection, retention, use, and sharing of customer data. This includes consent mechanisms, transparent privacy notices, and data minimisation principles. These policies should be reviewed annually and subject to independent audit.
- **Integrate ethical impact assessment into the project charter process.** Every project charter should include a mandatory section identifying potential harms to customers, employees, and communities. This embeds ethical scrutiny into the governance process from the outset, aligning with both deontological obligations and virtue ethics principles.
- **Formalise enterprise risk management aligned with COSO.** UPS should establish a risk register and regular internal audit cycle for IT governance, providing systematic oversight of emerging risks and ensuring that the governance framework adapts to the changing technology landscape.
- **Introduce innovation sandboxes to balance standardisation with agility.** As UPS continues to diversify, governance mechanisms should allow business units to pilot novel technologies in controlled environments without immediate requirement for full enterprise compliance. This preserves the benefits of standardisation while enabling strategic experimentation.

4. Conclusion

UPS's ICT governance transformation from the mid-1980s onwards provides a compelling example of how deliberate, structured governance can convert technology from a strategic liability into a competitive advantage. By establishing a layered decision-making framework spanning from the Steering Committee to business unit project teams, UPS ensured that IT investments remained aligned with corporate strategy, operationally disciplined, and responsive to emerging opportunities such as the internet era.

Benchmarked against COBIT, ITIL, ISO 27001, COSO, and ISO 9000, UPS's approach demonstrates genuine strength in governance structure and service reliability. The key gaps

lie in information security management and formalised risk oversight, which have grown more significant as data protection expectations and regulatory requirements have evolved.

Ethically, UPS's strategy broadly satisfies utilitarian objectives through wide-ranging value creation and deontological standards through transparent governance processes. However, the concentration of customer data in centralised systems creates ongoing privacy obligations that the documented strategy does not fully address. These risks could have been mitigated from the outset by incorporating privacy-by-design principles into architecture planning and establishing explicit data governance policies alongside the technical build-out. As UPS continues to expand into new markets and services, embedding data ethics and formal security governance at the core of its ICT strategy will be essential to maintaining the trust of customers, employees, and regulators alike.

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